





Quantum Optimal Control via Semi-Automatic Differentiation

Quantum 6, 871 (2022) — arXiv:2205.15044

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DEVCOM Army Research Lab

QP&ML Journal Club • December 13, 2022

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Quantum Optimal Control

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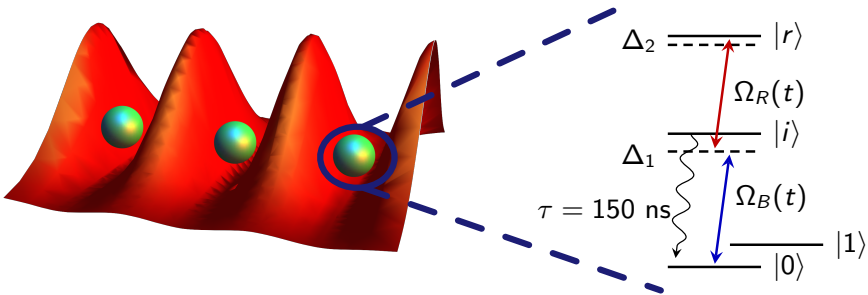
$$\hat{H} = \hat{H}(\{\epsilon_l(t)\}), \qquad \hat{H} = \hat{H}(\{\epsilon_n\})$$

What:

- State preparation
- Controlling photochemical processes
- Gate optimization
- ...

Optimal Control Tasks — Quantum Gates

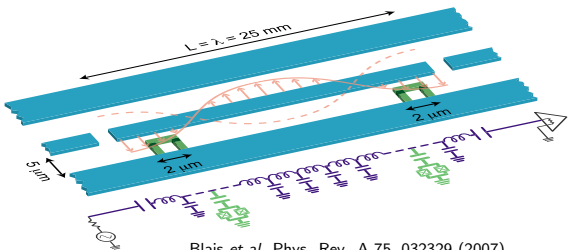
Rydberg atoms



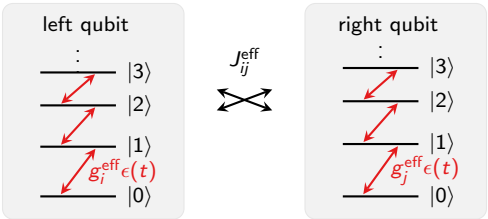
Goerz *et al.* Phys. Rev. A 90, 032329 (2014)

Optimal Control Tasks — Quantum Gates

Two transmon qubits with shared transmission line



Blais *et al.* Phys. Rev. A 75, 032329 (2007)



Goerz *et al.* EPJ Quantum Tech. 2, 21 (2015)
Goerz *et al.* npj Quantum Information 3, 37 (2017)

Quantum Control Problem

"Pulse-level" control

- Bunch of states: $\{|\Psi_k(t)\rangle\}$
  — e.g. two-qubit gate: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$

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  — assume piecewise-constant: ϵ_{ln} for n'th time interval of l'th control

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Functional

$$J(\{\epsilon_{nl}\}) = J_T(\{|\Psi_k(T)\rangle\}) + \int_0^T g_a\big(\{\epsilon_l(t)\}, t\big)\mathrm{d}t + \int_0^T g_b\big(\{|\Psi_k(t)\rangle\}, t\big)\mathrm{d}t$$

Quantum Control Problem

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Gradient-based "open loop" optimization

$$(\nabla J)_{ln} \equiv \frac{\partial J}{\partial \epsilon_{ln}} \qquad \Rightarrow \qquad \text{L-BFGS-B}$$

Quantum Control Problem

How to calculate the gradient $\frac{\partial J}{\partial \epsilon_{ln}}$?

Automatic Differentiation (AD)

PHYSICAL REVIEW A **95**, 042318 (2017)

Speedup for quantum optimal control from automatic differentiation based on graphics processing units

Nelson Leung,[1,*] Mohamed Abdelhafez,[1] Jens Koch,[2] and David Schuster[1]

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PHYSICAL REVIEW A **101**, 022321 (2020)

Universal gates for protected superconducting qubits using optimal control

Mohamed Abdelhafez,[1] Brian Baker,[2] András Gyenis,[3] Pranav Mundada,[3] Andrew A. Houck,[3] David Schuster,[1] and Jens Koch[2]

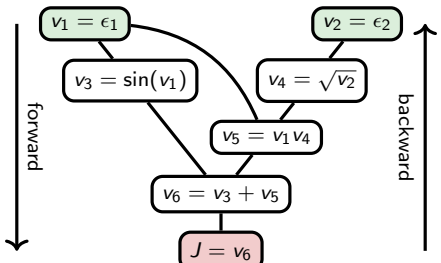
Automatic Differentiation (AD)

backward-mode "adjoint"

$$\bar{v}_j \equiv \frac{\partial J}{\partial v_j} = \sum_i \bar{v}_i \frac{\partial v_i}{\partial v_j}$$

sum over all v_i

which depend on v_j



Automatic Differentiation (AD)

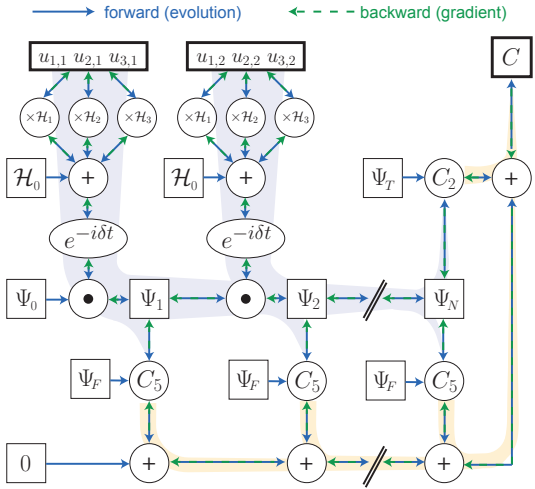


Fig. 2 in Leung *et al.* Phys. Rev. A 95, 042318 (2017)

AD Advantages

- Arbitrary functionals

μ	Cost-function contribution	$C_\mu(\mathbf{u})$
1	Target-gate infidelity	$1 - \lvert \mathrm{tr}(K_T^\dagger K_N)/D \rvert^2$
2	Target-state infidelity	$1 - \lvert \langle \Psi_T \vert \Psi_N \rangle \rvert^2$
3	Control amplitudes	$\lvert \mathbf{u} \rvert^2$
4	Control variations	$\sum_{j,k} \lvert u_{k,j} - u_{k,j-1} \rvert^2$
5	Occupation of forbidden state	$\sum_j \lvert \langle \Psi_F \vert \Psi_j \rangle \rvert^2$
6	Evolution time (target gate)	$1 - \frac{1}{N} \sum_j \lvert \mathrm{tr}(K_T^\dagger K_j)/D \rvert^2$
7	Evolution time (target state)	$1 - \frac{1}{N} \sum_j \lvert \langle \Psi_T \vert \Psi_j \rangle \rvert^2$

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- Arbitrary equations of motion
  e.g., quantum trajectories — Abdelhafez *et al.* Phys. Rev. A 99, 052327 (2019)

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- Arbitrary equations of motion
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- GPU support

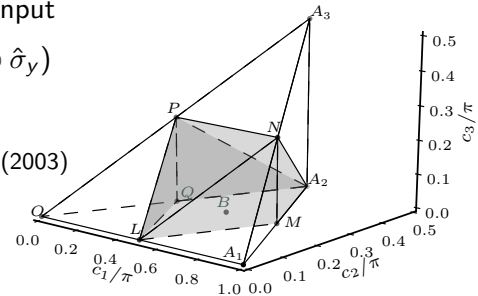
Entanglement Measures

Quantum Gate Concurrence

Max concurrence that can be generated for a separable input

1. $c_1, c_2, c_3 \propto \text{eigvals}\left(\hat{U}\tilde{U}\right)$; $\tilde{U} = (\hat{\sigma}_y \otimes \hat{\sigma}_y)\,\hat{U}\,(\hat{\sigma}_y \otimes \hat{\sigma}_y)$
2. $C(\hat{U}) = \max|\sin(c_{1,2,3} \pm c_{3,1,2})|$

Childs *et al.* Phys. Rev. A 68, 052311 (2003)



Entanglement Measures

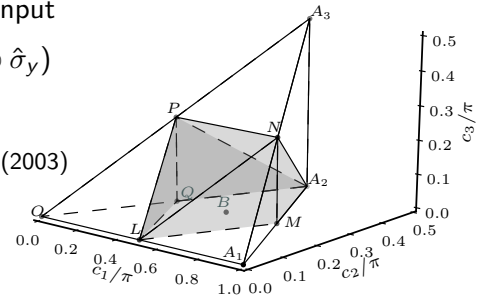
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Not analytic!



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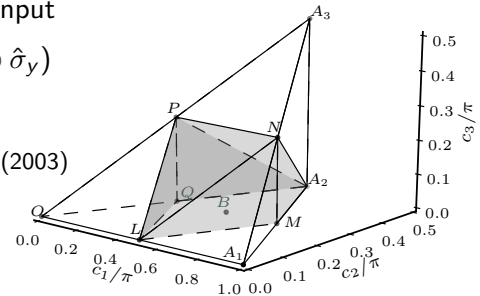
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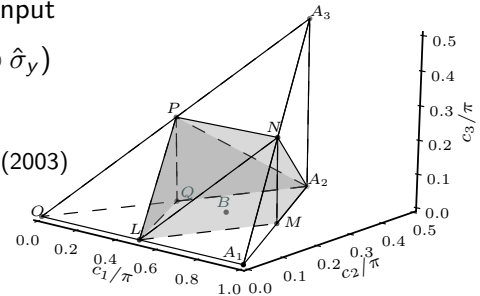
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Quantum Fisher Information

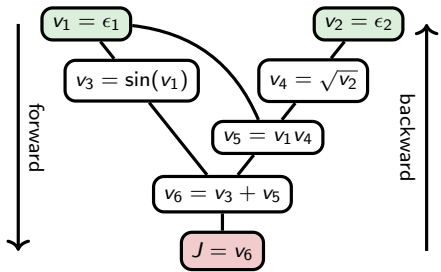
$$F(\hat{\rho}) = \sum_{i \neq j} \frac{2(p_i - p_j)^2}{p_i + p_j} \left| \left\langle \phi_i \middle| \hat{S}_z \middle| \phi_j \right\rangle \right|^2$$

where p_i, $|\phi_i\rangle$ are eigenvalues / eigenstates of $\hat{\rho}$

— Ma *et al.*. Phys. Rep. 509, 89 (2011)

AD Compromises

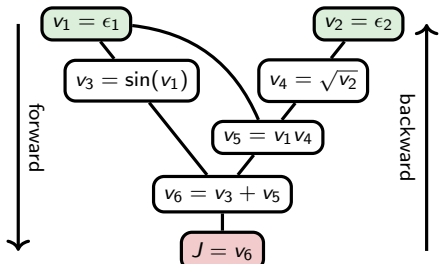
1. Numerical scaling
   - AD memory overhead
   - computational overhead (at least on CPU)



$$J(\epsilon_1, \epsilon_2) = \sin(\epsilon_1) + \epsilon_1\sqrt{\epsilon_2}$$

AD Compromises

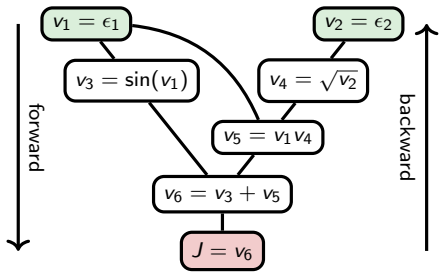
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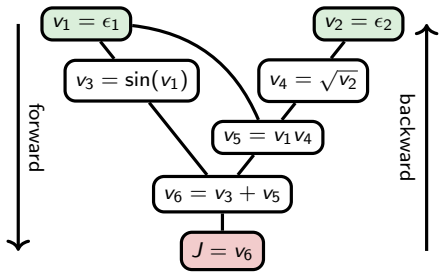
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We don't have to compromise!

Semi-Automatic Differentiation

Quantum 6, 871 (2022) — arXiv:2205.15044

Quantum Optimal Control via Semi-Automatic Differentiation

Michael H. Goerz, Sebastián C. Carrasco, and Vladimir S. Malinovsky

DEVCOM Army Research Laboratory, 2800 Powder Mill Road, Adelphi, MD 20783, USA



uant-ph] 1 Dec 2022

We develop a framework of "semi-automatic differentiation" that combines existing gradient-based methods of quantum optimal control with automatic differentiation. The approach allows to optimize practically any computable functional and is implemented in two open source Julia packages, `GRAPE.jl` and `Krotov.jl`, part of the `QuantumControl.jl` framework. Our method is based on formally rewriting the optimization functional in terms of propagated states, overlaps with target states, or quantum gates. An analytical application of the chain rule then allows to separate the time propagation and the evaluation of the functional when calculating the gradient. The former can be evaluated with great efficiency via a modified GRAPE scheme. The latter is evaluated with automatic differenti-

Funding

DEVCOM Army Research Laboratory, Cooperative Agreement Numbers W911NF-16-2-0147, W911NF-21-2-0037; DIRA-TRC No. DTR19-CI-019

arXiv:2205.15044

Semi-Automatic Differentiation

arXiv:2205.15044

Semi-Automatic Differentiation

$$J(\{\varepsilon_{nl}\}) = J_T(\{|\psi_k(T)\rangle\}) + \dots$$

$$\nabla J = \frac{\partial J_T}{\partial \varepsilon_{nl}}$$

$$\frac{\partial J_T}{\partial \varepsilon_{nl}} = 2\,\mathrm{Re}\left[\sum_k \underbrace{\frac{\partial J_T}{\partial \langle\psi_k(T)|}}_{|\chi_k(T)\rangle} \cdot \frac{\partial |\psi_k(T)\rangle}{\varepsilon_{nl}}\right]; \quad |\chi_k(T)\rangle = \frac{\partial J_T}{\partial \langle\psi_k(T)|}$$

$$= 2\,\mathrm{Re}\left[\sum_k \frac{\partial}{\partial \varepsilon_{nl}} \langle\chi_k(T)|\psi_k(T)\rangle\right]$$

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Semi-Automatic Differentiation

$$\frac{\partial}{\partial \varepsilon_{ne}} \langle \chi_k(T) | \hat{U}_N \cdots \hat{U}_1 | \psi_k(0) \rangle \qquad ; \hat{U}_n = e^{-i \hat{H}_n dt}$$

$$= \underbrace{\langle \chi_k(T) | \hat{U}_N \cdots \hat{U}_{n+1}}_{\text{bw}} \frac{\partial \hat{U}_n}{\partial \varepsilon_{ne}} \underbrace{\hat{U}_{n-1} \cdots \hat{U}_1 | \psi_k(0) \rangle}_{\text{fw}}$$

$$|\chi_k(T)\rangle = \frac{\partial J_T}{\partial \langle \psi_k(T)|}$$

$J_T(\hat{A})$ $\qquad \hat{A}_{ik} = \langle \phi_i | \psi_k \rangle$

$$\frac{\partial J_T}{\partial \langle \psi_k(T)|} = \sum_i \frac{\partial J_T}{\partial u_{ik}} |\phi_i\rangle$$

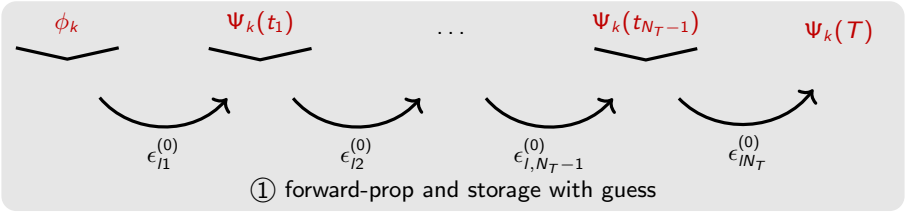
Gradient of Time Evolution Operator

$$\begin{pmatrix} \frac{\partial \hat{U}_n^\dagger}{\partial \epsilon_{n1}} |\chi_k(t_n)\rangle \\ \vdots \\ \frac{\partial \hat{U}_n^\dagger}{\partial \epsilon_{nL}} |\chi_k(t_n)\rangle \\ \hat{U}_n^\dagger |\chi_k(t_n)\rangle \end{pmatrix} = \exp\left[-\mathrm{i} \begin{pmatrix} \hat{H}_n^\dagger & 0 & \dots & 0 & \hat{H}_n^{(1)\dagger} \\ 0 & \hat{H}_n^\dagger & \dots & 0 & \hat{H}_n^{(2)\dagger} \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \dots & \hat{H}_n^\dagger & \hat{H}_n^{(L)\dagger} \\ 0 & 0 & \dots & 0 & \hat{H}_n^\dagger \end{pmatrix} dt_n \right] \begin{pmatrix} 0 \\ \vdots \\ 0 \\ |\chi_k(t_n)\rangle \end{pmatrix}$$

$$\hat{U}_n = \exp[-\mathrm{i}\hat{H}_n dt_n]\,; \qquad \hat{H}_n^{(l)} = \frac{\partial \hat{H}_n}{\partial \epsilon_l(t)}$$

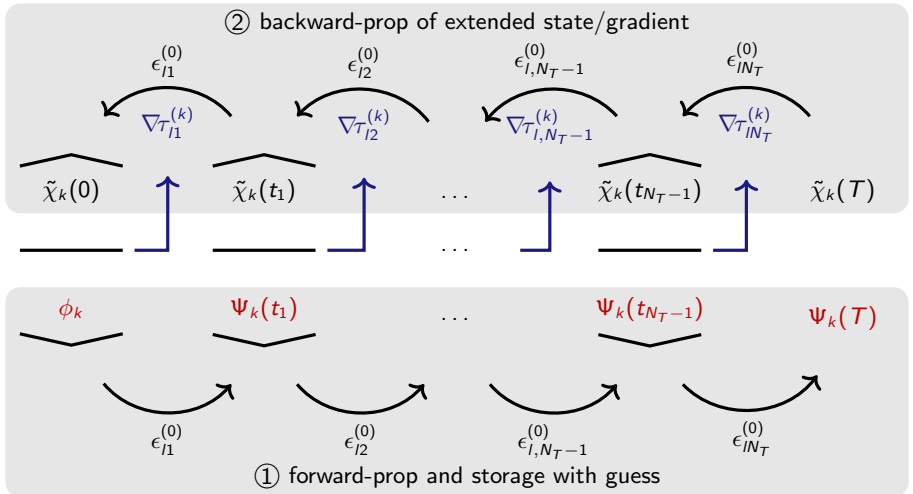
— Goodwin, Kuprov, J. Chem. Phys. 143, 084113 (2015)

Generalized GRAPE scheme

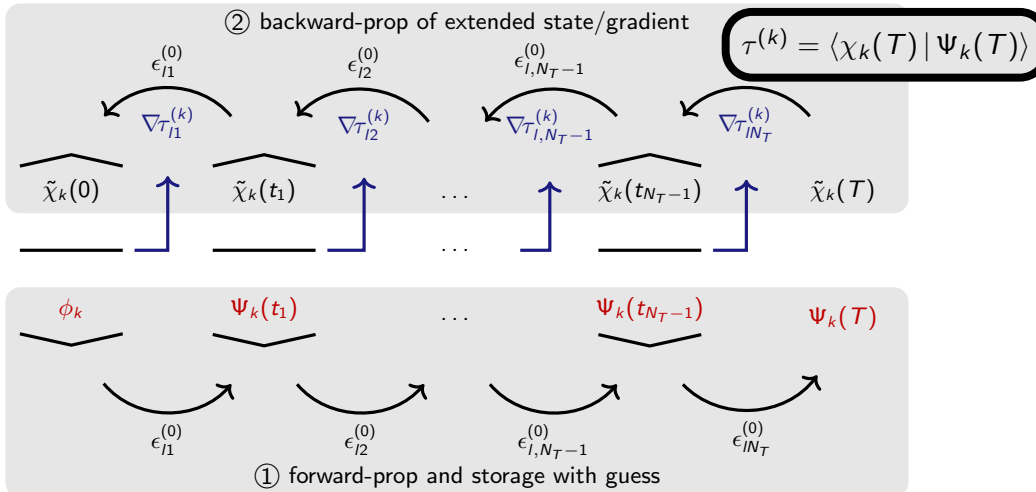


① forward-prop and storage with guess

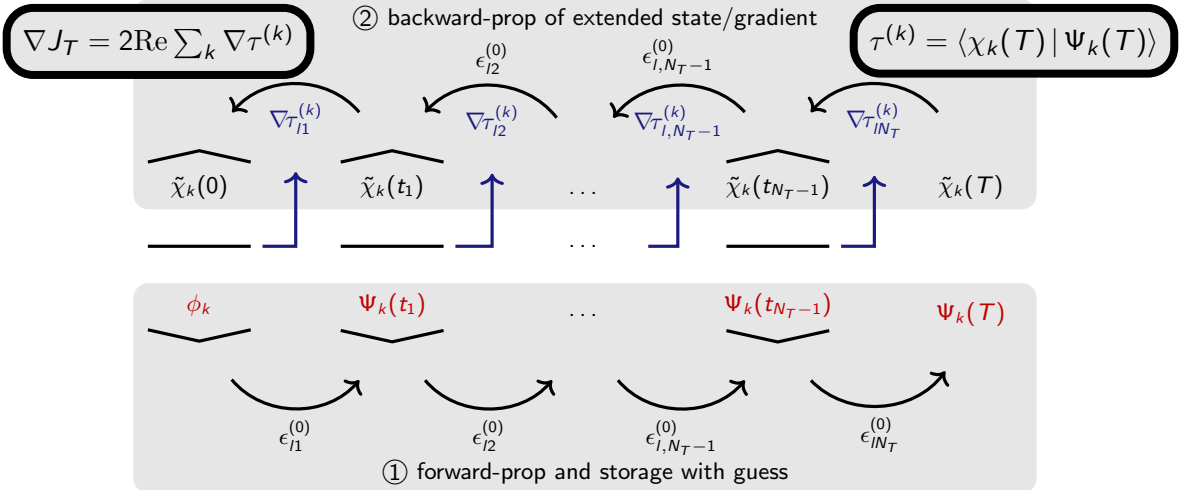
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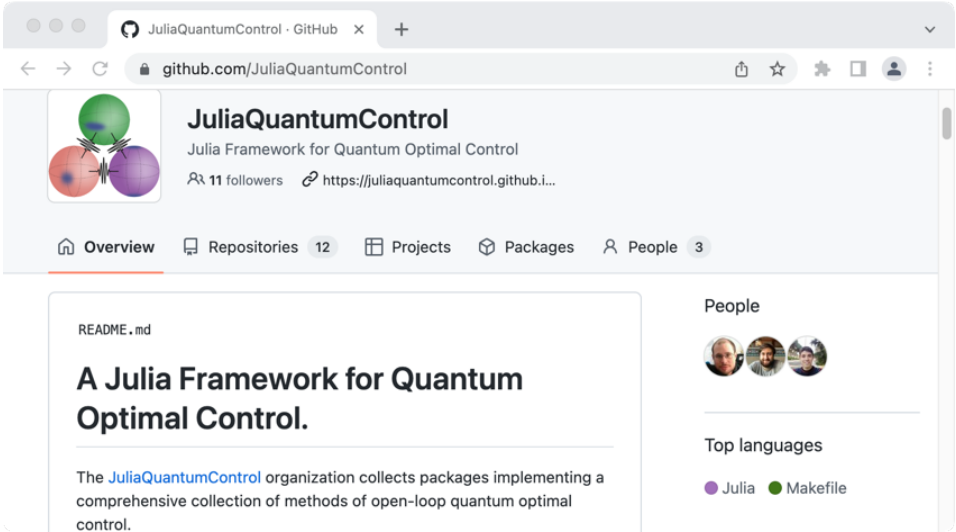
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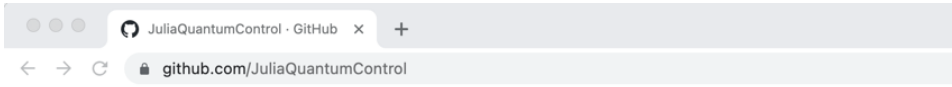
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JuliaQuantumControl



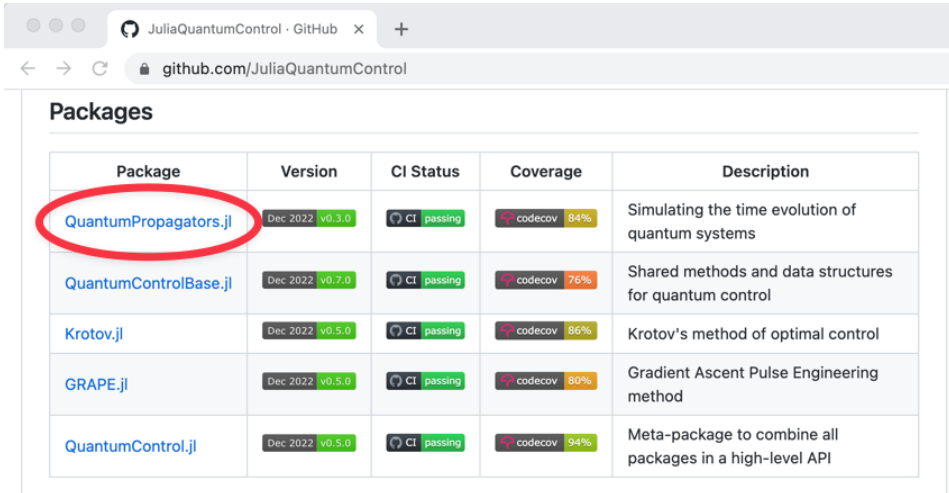
JuliaQuantumControl



Packages

Package	Version	CI Status	Coverage	Description
QuantumPropagators.jl	Dec 2022 v0.3.0	CI passing	codecov 84%	Simulating the time evolution of quantum systems
QuantumControlBase.jl	Dec 2022 v0.7.0	CI passing	codecov 76%	Shared methods and data structures for quantum control
Krotov.jl	Dec 2022 v0.5.0	CI passing	codecov 86%	Krotov's method of optimal control
GRAPE.jl	Dec 2022 v0.5.0	CI passing	codecov 80%	Gradient Ascent Pulse Engineering method
QuantumControl.jl	Dec 2022 v0.5.0	CI passing	codecov 94%	Meta-package to combine all packages in a high-level API

JuliaQuantumControl



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QuantumControl.jl	Dec 2022 v0.5.0	CI passing	codecov 94%	Meta-package to combine all packages in a high-level API

Propagation Methods

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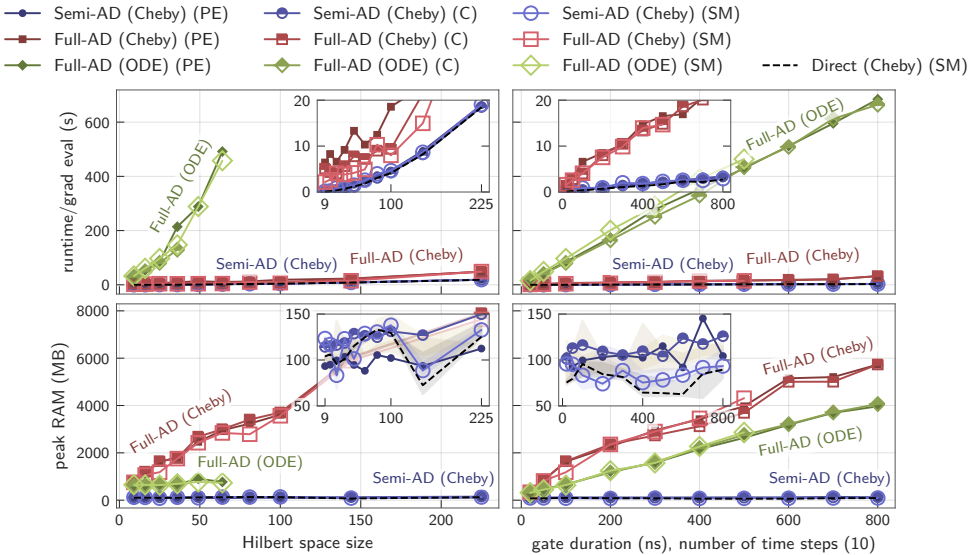
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  - Faster converging polynomials $\Rightarrow$ Chebychev

Demo

Benchmarks


Semi-AD (Cheby) (PE)
Full-AD (Cheby) (PE)
Full-AD (ODE) (PE)
Semi-AD (Cheby) (C)
Full-AD (Cheby) (C)
Full-AD (ODE) (C)
Semi-AD (Cheby) (SM)
Full-AD (Cheby) (SM)
Full-AD (ODE) (SM)
Direct (Cheby) (SM)
runtime/grad eval (s)
peak RAM (MB)
Full-AD (ODE)
Semi-AD (Cheby)
Full-AD (Cheby)
Hilbert space size
gate duration (ns), number of time steps (10)

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AD-enhanced optimal control without compromises! arXiv:2205.15044

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Thank you